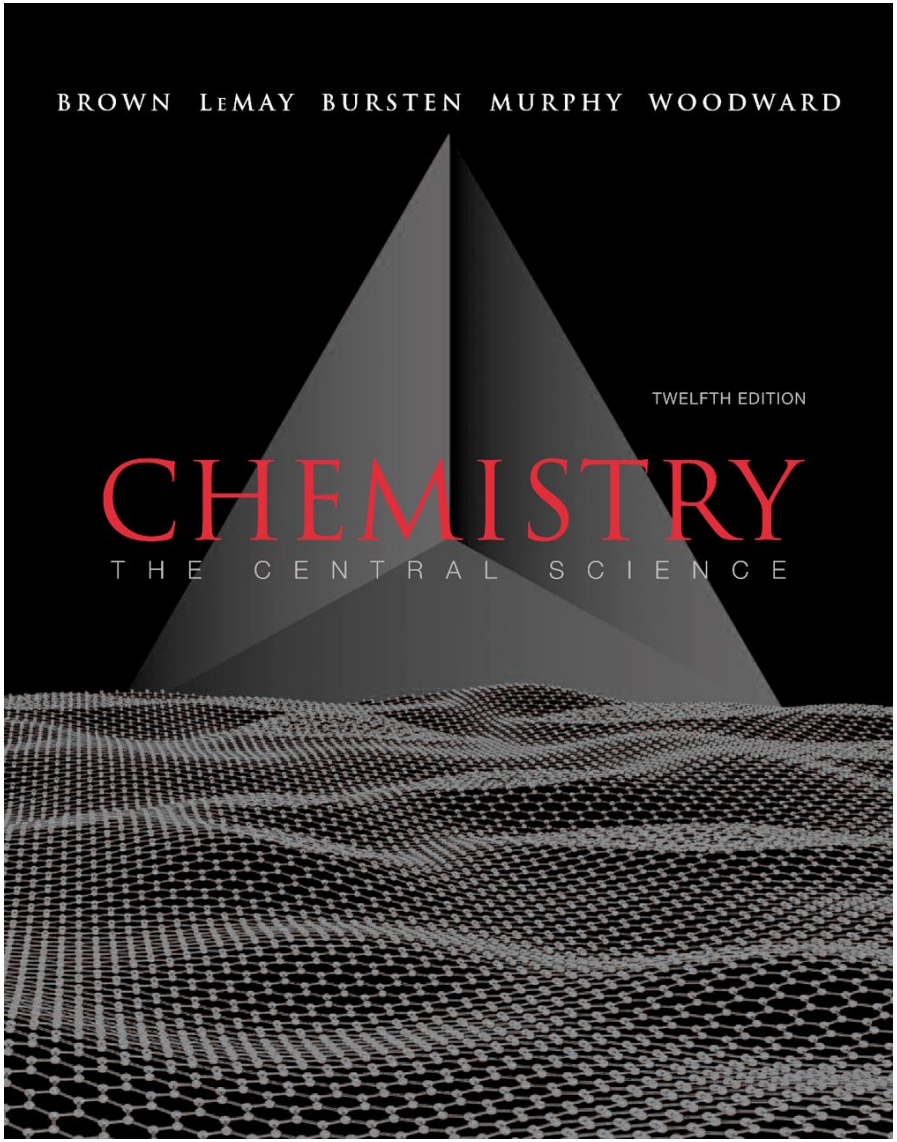


Lecture Presentation

Chapter 3

Stoichiometry: Calculations with Chemical Formulas and Equations

John D. Bookstaver
St. Charles Community College
Cottleville, MO

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Law of Conservation of Mass

"We may lay it down as an incontestable axiom that, in all the operations of art and nature, nothing is created; an equal amount of matter exists both before and after the experiment. Upon this principle, the whole art of performing chemical experiments depends."

--Antoine Lavoisier, 1789

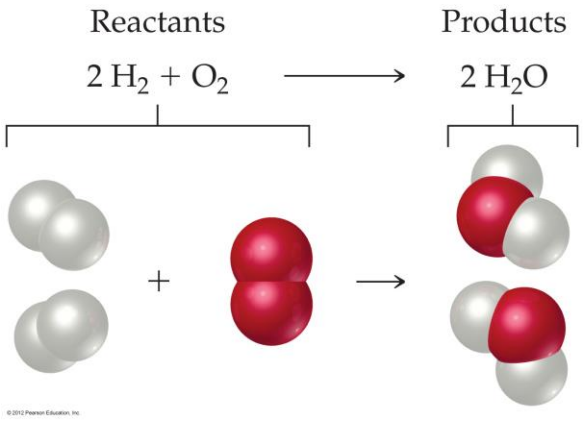


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Chemical Equations

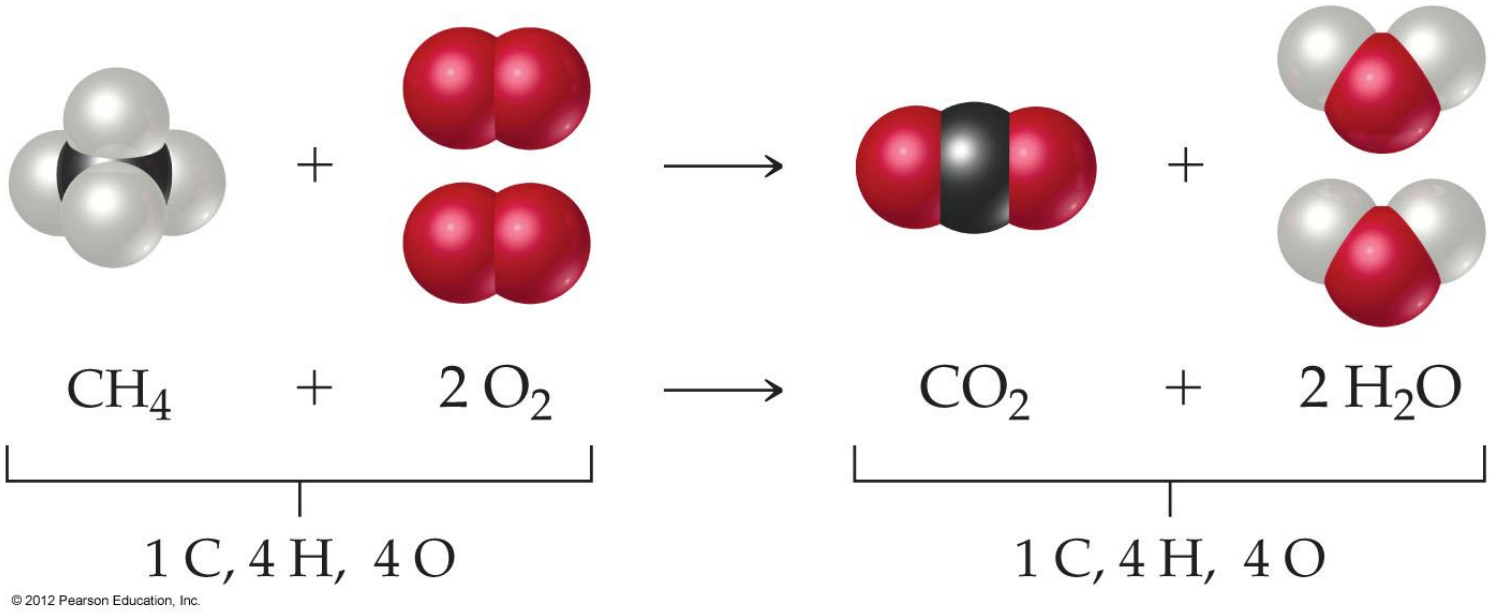
Chemical equations are concise representations of chemical reactions.



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Anatomy of a Chemical Equation

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$$

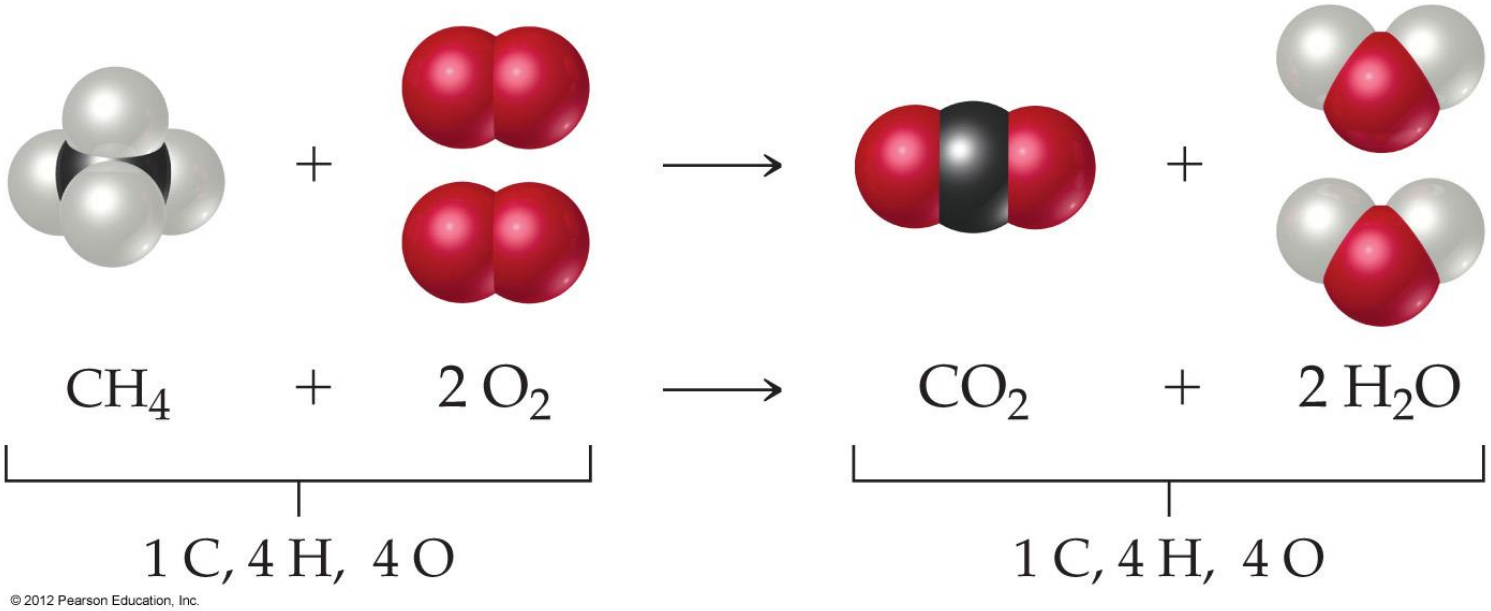


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Anatomy of a Chemical Equation

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$$



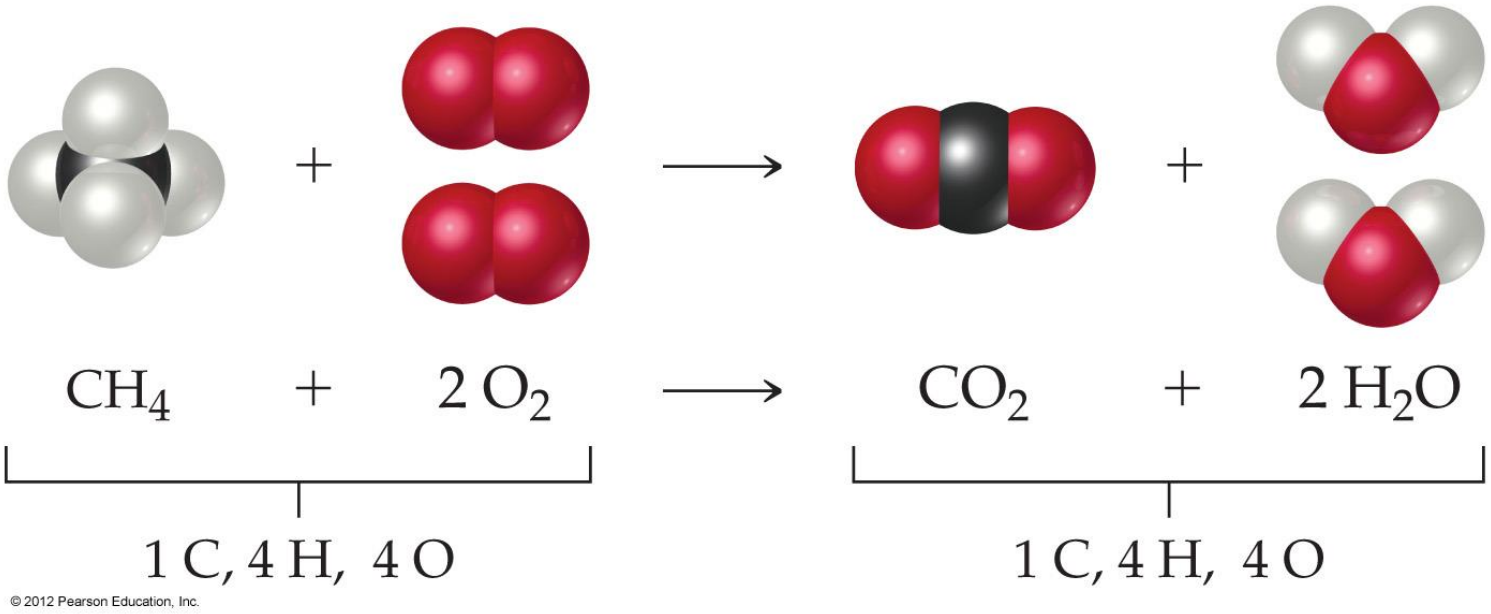
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Reactants appear on the left side of the equation.

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Anatomy of a Chemical Equation

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$$



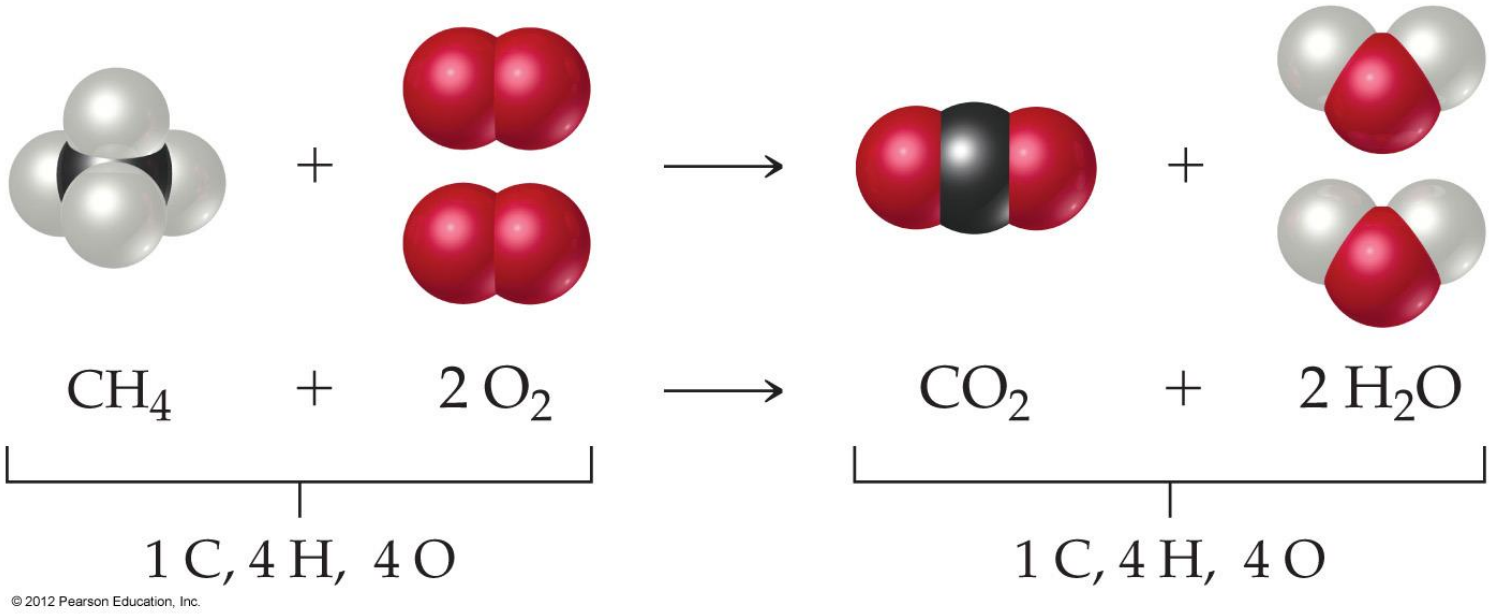
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Products appear on the right side of the equation.

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Anatomy of a Chemical Equation

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$$



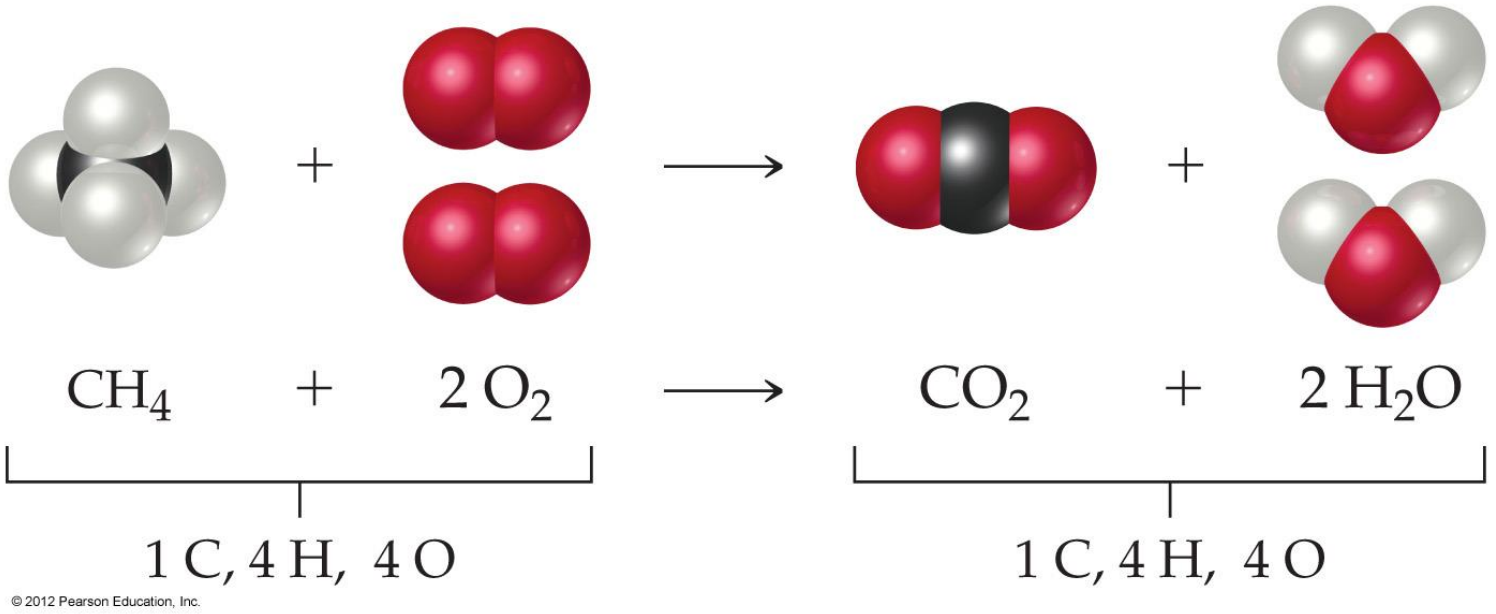
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The **states** of the reactants and products are written in parentheses to the right of each compound.

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Anatomy of a Chemical Equation

$$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$$

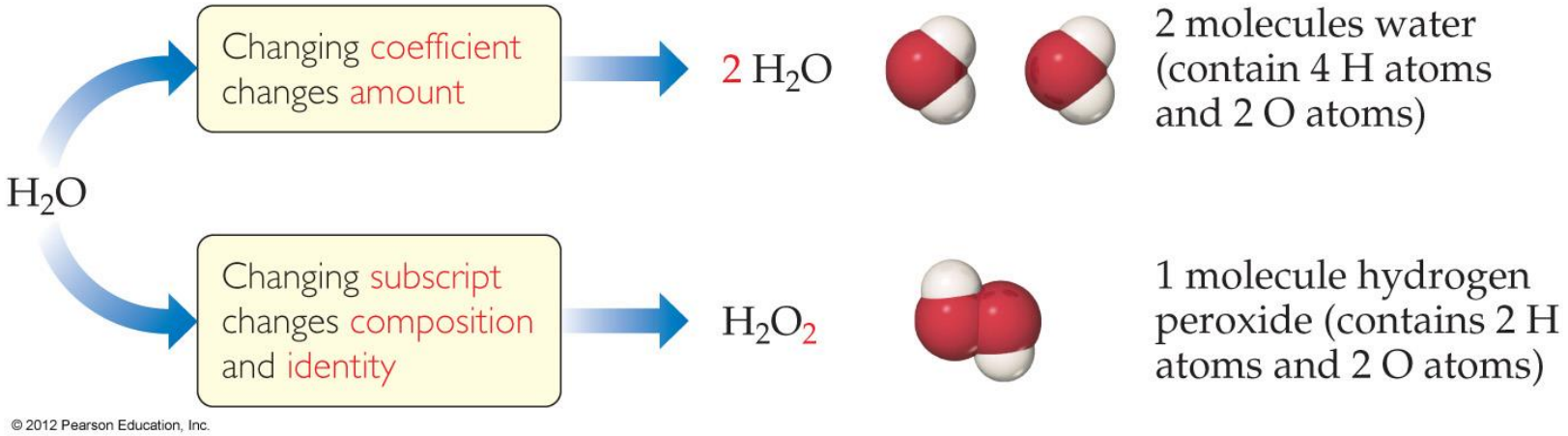


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Coefficients are inserted to balance the equation.

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Subscripts and Coefficients Give Different Information



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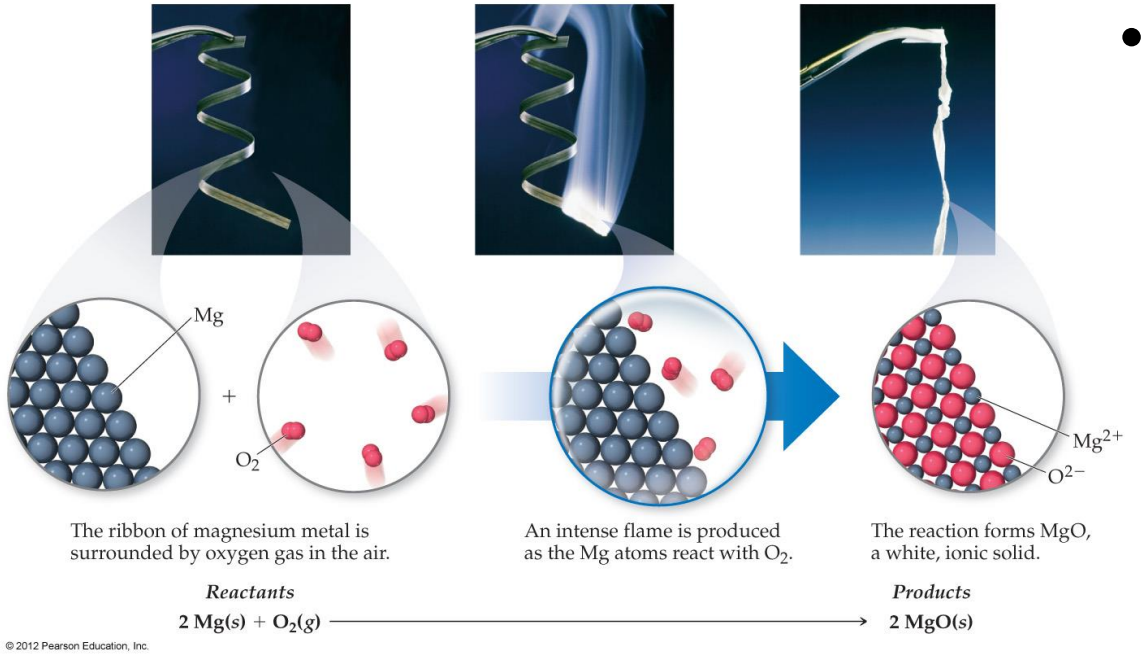
- **Subscripts** tell the number of atoms of each element in a molecule.
- **Coefficients** tell the number of molecules.

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Reaction Types

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Combination Reactions



- In **combination reactions** two or more substances react to form one product.

- Examples:
  - $2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$
  - $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$
  - $C_3H_6(g) + Br_2(l) \longrightarrow C_3H_6Br_2(l)$

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Decomposition Reactions

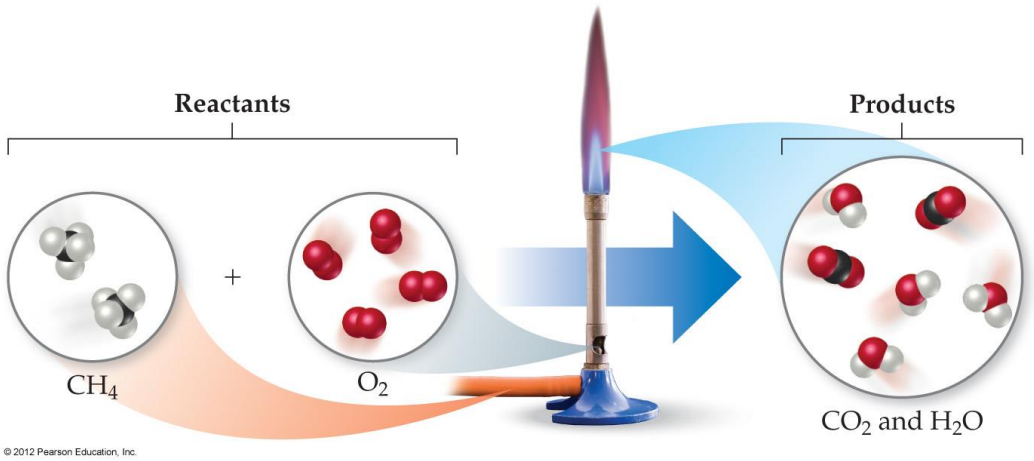


- In a **decomposition reaction** one substance breaks down into two or more substances.

- Examples:
  - $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$
  - $2KClO_3(s) \longrightarrow 2KCl(s) + O_2(g)$
  - $2NaN_3(s) \longrightarrow 2Na(s) + 3N_2(g)$

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Combustion Reactions



- **Combustion reactions** are generally rapid reactions that produce a flame.
- **Combustion reactions** most often involve hydrocarbons reacting with oxygen in the air.

- Examples:
  - $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$
  - $C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(g)$

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Formula Weights

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Formula Weight (FW)

- A **formula weight** is the sum of the atomic weights for the atoms in a chemical formula.
- So, the formula weight of calcium chloride, $CaCl_2$, would be

Ca: 1(40.08 amu)
+ Cl: 2(35.453 amu)
110.99 amu

- Formula weights are generally reported for ionic compounds.

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Molecular Weight (MW)

- A **molecular weight** is the sum of the atomic weights of the atoms in a molecule.
- For the molecule ethane, C_2H_6, the molecular weight would be

C: 2(12.011 amu)
+ H: 6(1.00794 amu)
30.070 amu

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Percent Composition

One can find the percentage of the mass of a compound that comes from each of the elements in the compound by using this equation:

$$\%\ \text{Element} = \frac{(\text{number of atoms})(\text{atomic weight})}{(\text{FW of the compound})} \times 100$$

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Percent Composition

So the percentage of carbon in ethane is

$$\%C = \frac{(2)(12.011\ \text{amu})}{(30.070\ \text{amu})}$$

$$= \frac{24.022\ \text{amu}}{30.070\ \text{amu}} \times 100$$

$$= 79.887\%$$

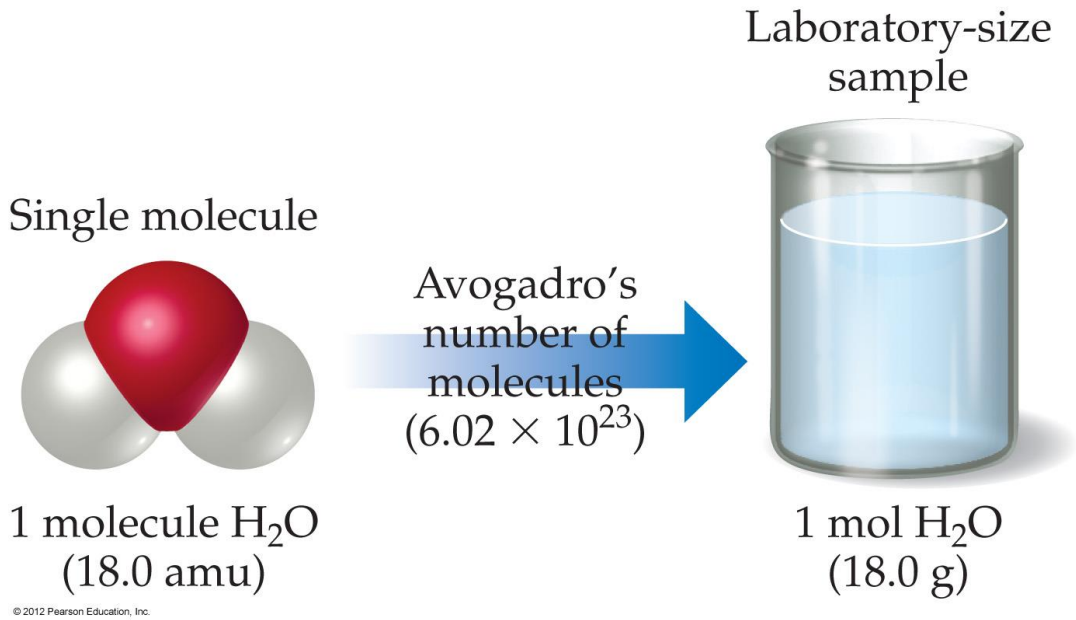
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Moles



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Avogadro's Number



- 6.02×10^{23}
- 1 mole of ^{12}C has a mass of 12.000 g.

Stoichiometry

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Molar Mass

- By definition, a **molar mass** is the mass of 1 mol of a substance (i.e., g/mol).
  - The molar mass of an element is the mass number for the element that we find on the periodic table.
  - The formula weight (in amu's) will be the same number as the molar mass (in g/mol).

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Using Moles



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Moles provide a bridge from the molecular scale to the real-world scale.

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Mole Relationships

TABLE 3.2 • Mole Relationships

Name of Substance	Formula	Formula Weight (amu)	Molar Mass (g/mol)	Number and Kind of Particles in One Mole
Atomic nitrogen	N	14.0	14.0	6.02×10^{23} N atoms
Molecular nitrogen	N_2	28.0	28.0	6.02×10^{23} N_2 molecules $2(6.02 \times 10^{23})$ N atoms
Silver	Ag	107.9	107.9	6.02×10^{23} Ag atoms
Silver ions	Ag^+	107.9*	107.9	6.02×10^{23} Ag^+ ions
Barium chloride	$BaCl_2$	208.2	208.2	6.02×10^{23} $BaCl_2$ formula units 6.02×10^{23} Ba^{2+} ions $2(6.02 \times 10^{23})$ Cl^- ions

*Recall that the electron has negligible mass; thus, ions and atoms have essentially the same mass.

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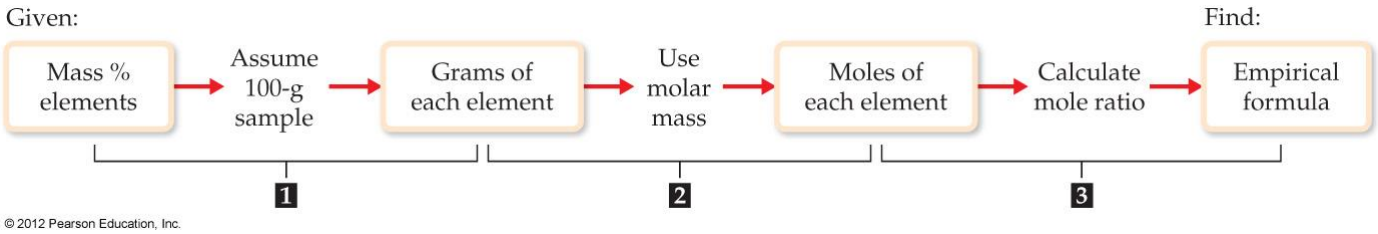
- One mole of atoms, ions, or molecules contains Avogadro's number of those particles.
- One mole of molecules or formula units contains Avogadro's number times the number of atoms or ions of each element in the compound.

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Finding Empirical Formulas

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Calculating Empirical Formulas



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One can calculate the empirical formula from the percent composition.

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Calculating Empirical Formulas

The compound *para*-aminobenzoic acid (you may have seen it listed as PABA on your bottle of sunscreen) is composed of carbon (61.31%), hydrogen (5.14%), nitrogen (10.21%), and oxygen (23.33%). Find the empirical formula of PABA.



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Calculating Empirical Formulas

Assuming 100.00 g of *para*-aminobenzoic acid,

$$\text{C:} \quad 61.31\ \text{g} \times \frac{1\ \text{mol}}{12.01\ \text{g}} = 5.105\ \text{mol C}$$

$$\text{H:} \quad 5.14\ \text{g} \times \frac{1\ \text{mol}}{1.01\ \text{g}} = 5.09\ \text{mol H}$$

$$\text{N:} \quad 10.21\ \text{g} \times \frac{1\ \text{mol}}{14.01\ \text{g}} = 0.7288\ \text{mol N}$$

$$\text{O:} \quad 23.33\ \text{g} \times \frac{1\ \text{mol}}{16.00\ \text{g}} = 1.456\ \text{mol O}$$

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Calculating Empirical Formulas

Calculate the mole ratio by dividing by the smallest number of moles:

$$\text{C:} \quad \frac{5.105 \text{ mol}}{0.7288 \text{ mol}} = 7.005 \approx 7$$

$$\text{H:} \quad \frac{5.09 \text{ mol}}{0.7288 \text{ mol}} = 6.984 \approx 7$$

$$\text{N:} \quad \frac{0.7288 \text{ mol}}{0.7288 \text{ mol}} = 1.000$$

$$\text{O:} \quad \frac{1.458 \text{ mol}}{0.7288 \text{ mol}} = 2.001 \approx 2$$



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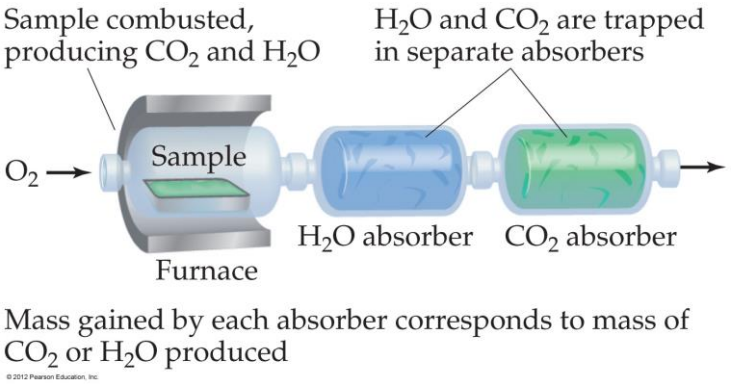
Calculating Empirical Formulas

These are the subscripts for the empirical formula:

$C_7H_7NO_2$

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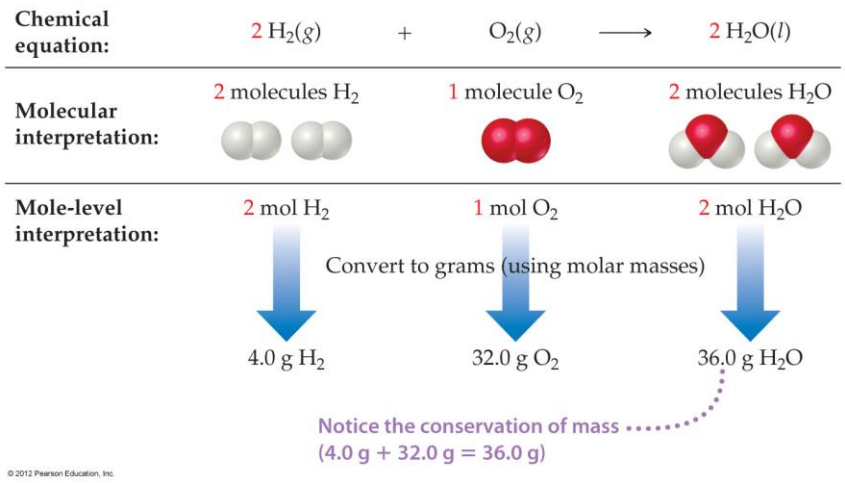
Combustion Analysis



- Compounds containing C, H, and O are routinely analyzed through combustion in a chamber like the one shown in Figure 3.14.
  - C is determined from the mass of CO_2 produced.
  - H is determined from the mass of H_2O produced.
  - O is determined by difference after the C and H have been determined.

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Stoichiometric Calculations

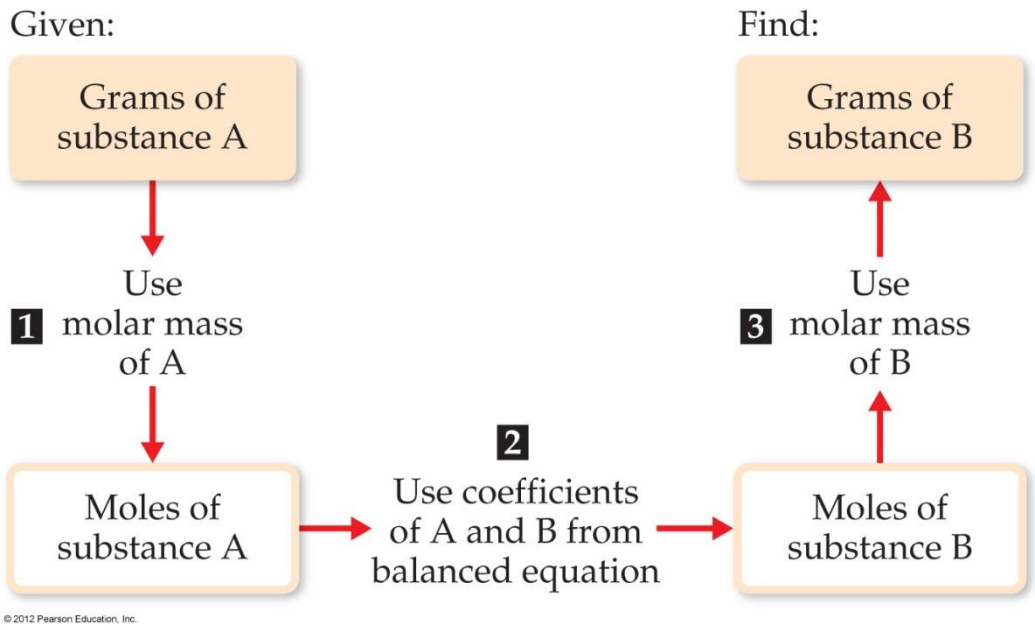


The coefficients in the balanced equation give the ratio of *moles* of reactants and products.

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Stoichiometric Calculations

Starting with the mass of Substance A, you can use the ratio of the coefficients of A and B to calculate the mass of Substance B formed (if it's a product) or used (if it's a reactant).

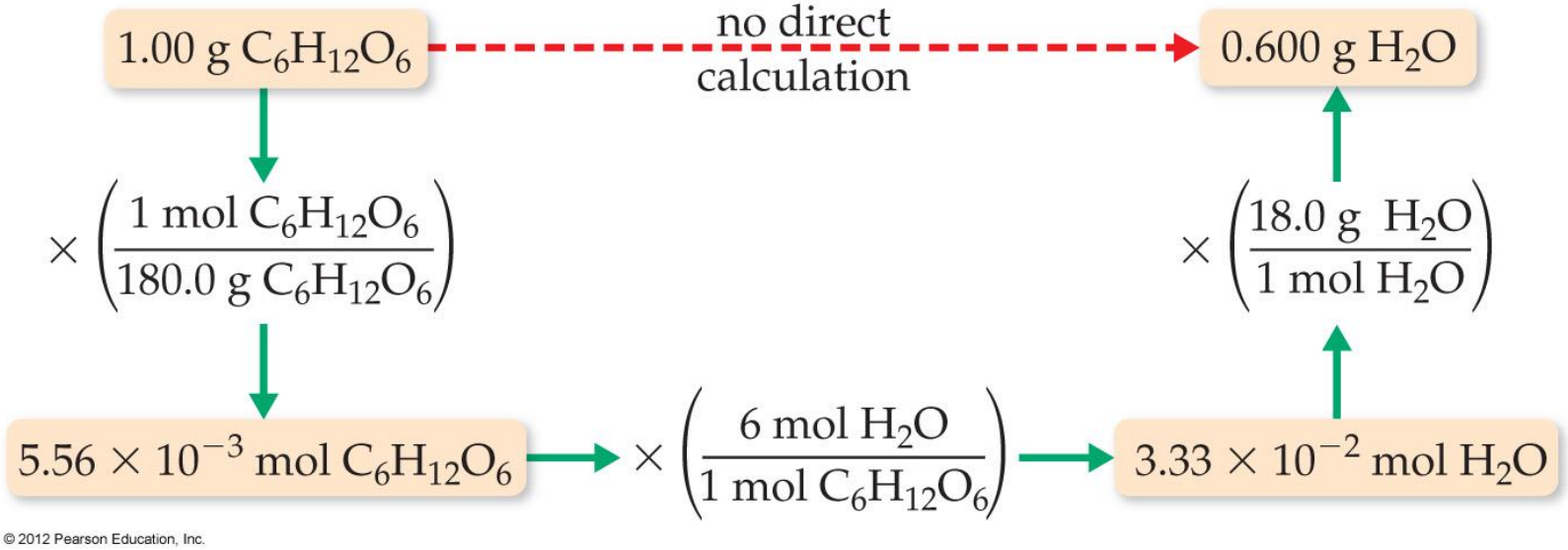


Stoichiometry

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Stoichiometric Calculations

$$C_6H_{12}O_6 + 6\ O_2 \longrightarrow 6\ CO_2 + 6\ H_2O$$



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Starting with 1.00 g of $C_6H_{12}O_6$…
we calculate the moles of $C_6H_{12}O_6$…
use the coefficients to find the moles of H_2O…
and then turn the moles of water to grams.

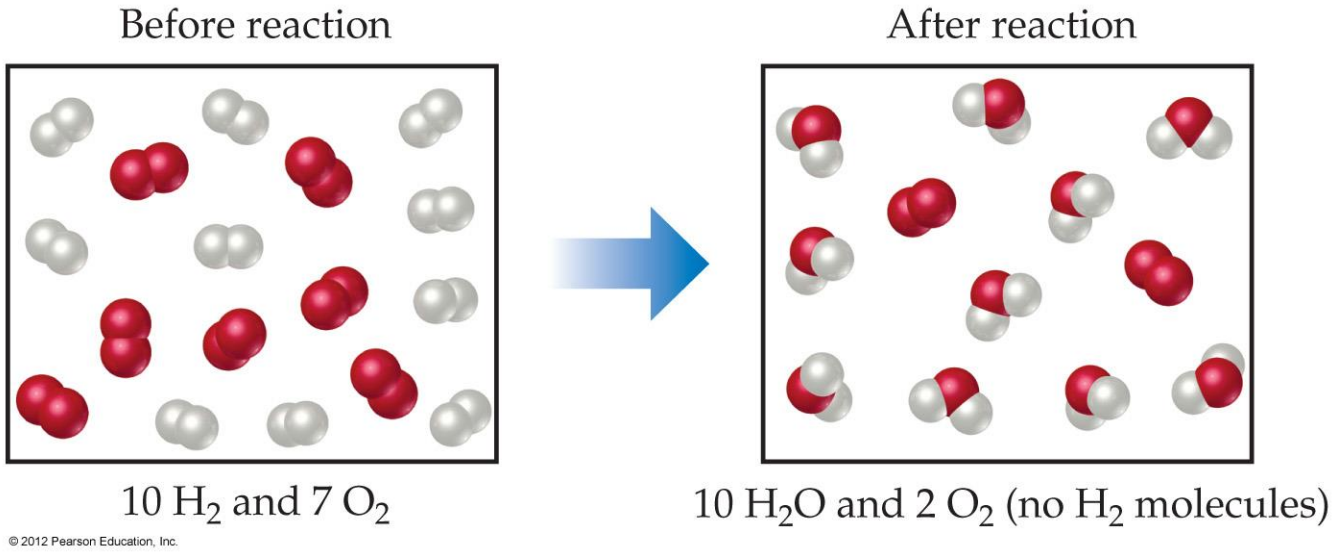
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Limiting Reactants

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Limiting Reactants

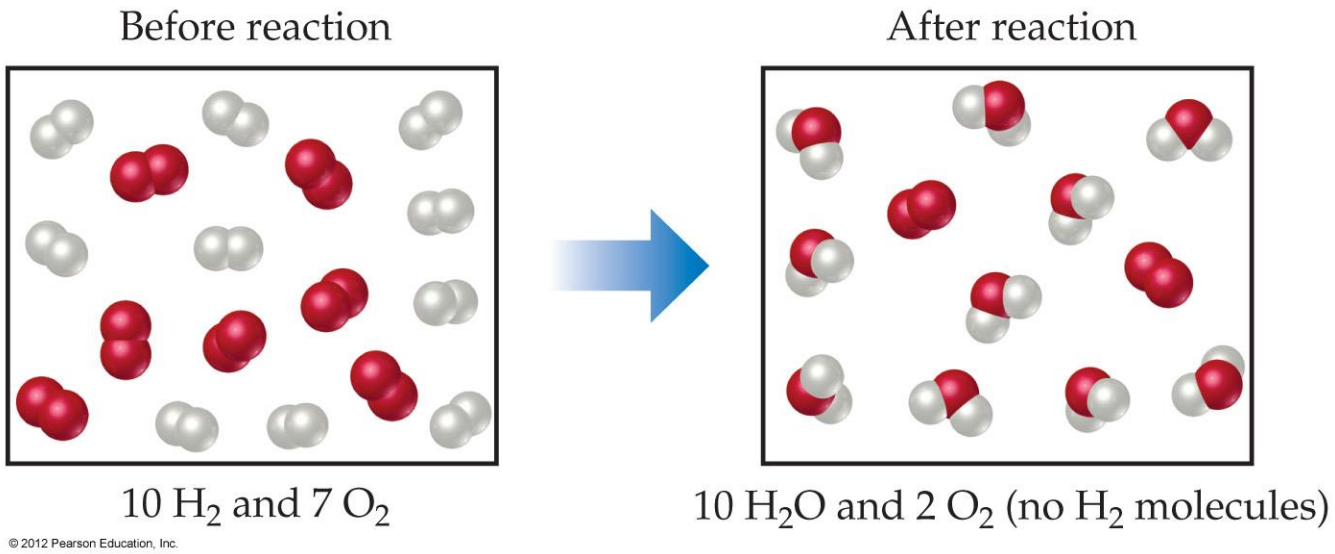
- The **limiting reactant** is the reactant present in the smallest stoichiometric amount.
  - In other words, it's the reactant you'll run out of first (in this case, the H_2).



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Limiting Reactants

In the example below, the O_2 would be the **excess reagent**.



Stoichiometry

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Theoretical Yield

- The **theoretical yield** is the maximum amount of product that can be made.
  - In other words, it's the amount of product possible as calculated through the stoichiometry problem.
- This is different from the **actual yield,** which is the amount one actually produces and measures.

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Percent Yield

One finds the **percent yield** by comparing the amount actually obtained (actual yield) to the amount it was possible to make (theoretical yield):

$$\text{Percent yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$



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